

General diagonalization for complex matrices

Theorem

If A is a complex $n \times n$ and A has n eigenvalues (counting multiplicities) then A is diagonalizable iff for every eigenvalue, the algebraic multiplicity equals the geometric multiplicity i.e. for every eigenvalue λ , the algebraic multiplicity of λ equals the dimension of the λ -eigenspace.

Ex 1

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

- eigenvalue 1 has alg. mult 2.
- 1-eigenspace is just 1-dim
line, for $z \in \mathbb{C}$.

Ex: $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

eigenvalues: $0, 2$
 $\uparrow$ $\uparrow$
 alg. mult. 1 $\uparrow$ alg. mult. 2.
alg. mult. 2.

2-eigenspace: $\underline{s \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}$, $s, t \in \mathbb{C}$.

dim of this eigenspace is 2.

0-eigenspace: $\underline{z \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}$, dim of the 0-eigenspace is 1.

A is diagonalizable and if $D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

and $P^{-1}AP = D$ where $P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$

invert.
because
the columns
are lin. indep.

basis for
the 2-eigen.

0-eigen.

Step-by-step diagonalization algorithm

- 1 Suppose $\lambda_1, \dots, \lambda_k$ are the eigenvalues of A and their algebraic multiplicities are $n_1, \dots, n_k$.
- 2 For each $i = 1, \dots, k$, determine a basis B_i for the λ_i -eigenspace of A . If for any i , the dimension of this eigenspace is less than n_i then A is not diagonalizable.
- 3 Otherwise, collect together all the bases, $B_1, \dots, B_k$ and place these vectors in the columns of a matrix P .
- 4 P will be invertible and $AP = PD$ where D is diagonal with diagonal entries $\lambda_1, \dots, \lambda_k$ repeated $n_1, \dots, n_k$ many times respectively.

Ex:

$$A = \left(\begin{array}{cc|cc} 1 & 0 & -1 & -1 \\ 1 & 3 & 1 & 0 \\ \hline 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right)$$

$$\begin{aligned} \det(A - \lambda I) &= (\lambda - 1)(\lambda - 3) ((2 - \lambda)^2 - 1) \\ &= (\lambda - 1)^2 (\lambda - 3)^2 \end{aligned}$$

eigenvalues: $\lambda = 1, 3$ and both have alg. mult 2

geom. mult of $\lambda = 1, 3$:

$$\lambda = 1: \text{ Solve } (A - I)u = 0$$

Solve $\begin{pmatrix} 0 & 0 & -1 & -1 \\ 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} u = 0$

The 1-eigenspace has dim. 2. The 1-eigenspace

$$u = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \quad s, t \in \mathbb{C}.$$

$\lambda = 3$: Solve

$$\begin{pmatrix} -2 & 0 & -1 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} u = 0$$

$$u = s \underbrace{\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}} + t \underbrace{\begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix}}, \quad s, t \in \mathbb{C}.$$

geom mult. is 2.

So the original A is diagonalizable.

$$P = \begin{pmatrix} -2 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \leftarrow \text{invert. and}$$

$\uparrow$
 basis for. 1-eigen. basis for 3-eigen.

$$P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

Ex:

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

char poly of A is $\det(A - \lambda I) = \det$

$$\begin{pmatrix} -\lambda & 0 & 0 & 1 \\ 1 & -\lambda & 0 & 0 \\ 0 & 1 & -\lambda & 0 \\ 0 & 0 & 1 & -\lambda \end{pmatrix}$$

14 spots

$$= (-\lambda)(-\lambda)^3 - 1$$

$$(-1)^{44} \cdot 1 \cdot \det(\sim)$$

$$= \underline{\lambda^4 - 1}$$

4 eigenvalues: $\lambda = \pm 1, \pm i$ so A is diag.

What are the eigenvectors?

$$\lambda = 1: \quad \text{Solve} \quad \begin{pmatrix} -1 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} u = 0$$

$$u = t \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad \text{for } t \in \mathbb{C}.$$

$$\lambda = -1: \quad \text{Solve} \quad \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} u = 0$$
$$u = t \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}$$

$$\lambda = i : \begin{pmatrix} -i & 0 & 0 & 1 \\ 1 & -i & 0 & 0 \\ 0 & 1 & -i & 0 \\ 0 & 0 & 1 & -i \end{pmatrix} u = 0$$

$$iu_1 = u_4$$

$$u_1 = iu_2$$

$$u_2 = iu_3$$

$$u_3 = iu_4$$

$$\text{so } u = \begin{pmatrix} -i \\ -1 \\ i \\ 1 \end{pmatrix}$$

$$\lambda = -i : \text{Check that } u = \begin{pmatrix} -i \\ 1 \\ -i \\ 1 \end{pmatrix} \text{ is the eigenspace.}$$

So A is diagonalized by

$$P = \begin{pmatrix} 1 & 1 & -i & -i \\ 1 & -1 & -1 & 1 \\ 1 & 1 & i & -i \\ 1 & -1 & 1 & 1 \end{pmatrix} \quad \leftarrow \text{invert.}$$

$$\text{and } P^{-1}AP = \begin{pmatrix} 1 & & 0 \\ & -1 & \\ 0 & & i \\ & & & -i \end{pmatrix}.$$

The general notion of an eigenvalue

Definition

Suppose that $T : V \rightarrow V$ is a linear transformation defined on a vector space V , If $T(v) = \lambda v$ for some $v \in V$ which is not zero we say that λ is an eigenvalue for T and v is an eigenvector for λ .

Matrix associated to a linear transformation

- 1 Suppose that V is finite-dimensional with basis $B = \{b_1, \dots, b_n\}$, and $T : V \rightarrow V$ is a linear transformation.
- 2 We can compute $T(b_1), \dots, T(b_n)$ and consider their B -coordinates.
- 3 If we let A be the matrix

$$\begin{bmatrix} T(b_1)_B & T(b_2)_B & \dots & T(b_n)_B \end{bmatrix}$$

then for any $v \in V$, $T(v)_B = Av_B$.

Change of basis matrix

- 1 Suppose we have a basis $B = \{b_1, \dots, b_n\}$ for $\mathbb{R}^n$.
- 2 Consider the identity map $id : \mathbb{R}^n \rightarrow \mathbb{R}^n$ where we wish to consider the domain written in standard coordinates and the codomain written in B coordinates.
- 3 The matrix for this transformation looks like

$$P = \begin{bmatrix} [e_1]_B & [e_2]_B & \dots & [e_n]_B \end{bmatrix}$$